

QTM 220 Final Practice Problems

Problem 1

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n+2} \sum_{i=1}^n Y_i$

Part A

What is $E[\tilde{Y}]$?

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$)

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

Part D

How does $Var[\tilde{Y}]$ compare to $Var[\bar{Y}]$?